



Redefining Polycystic Ovary Syndrome: Transformative Diagnostic and Management Changes in the 2023 Guidelines

A Ram Hong

Division of Endocrinology and Metabolism, Department of Internal Medicine, Chonnam National University Medical School, Gwangju, Korea

Polycystic ovary syndrome (PCOS) is an endocrine disorder characterized by hyperandrogenism (clinical and/or biochemical), ovulatory dysfunction, and polycystic ovarian morphology. PCOS affects between 10% and 13% of women of reproductive age [1,2]. In Korea, the age-adjusted incidence and prevalence of PCOS were reported as 2.8% and 4.3%, respectively, based on data from the National Health Information Database spanning 2010 to 2019 [3]. The diverse etiology and clinical manifestations of PCOS have contributed to delays in diagnosis and dissatisfaction with care. To improve understanding of PCOS and address gaps in practice, ‘The international evidence-based guideline for the assessment and management of polycystic ovary syndrome’ was initially published in 2018 [4]. A revised version of the guideline was released in 2023, following a comprehensive, evidence-based development process that included extensive engagement with healthcare professionals, as well as consumers and patients [5]. The aim of this guideline is to standardize care worldwide and promote high-quality clinical services to enhance health outcomes and quality of life for women with PCOS.

Several key recommendations and updates in the 2023 guideline are noteworthy from an endocrinologist’s perspective.

The most significant change in the 2023 guideline is the simplified diagnostic algorithm for PCOS. Generally, it follows the

2018 guideline, which is based on the 2003 Rotterdam criteria. These criteria state that a diagnosis requires the presence of any two of the following three factors: (1) clinical or biochemical hyperandrogenism, (2) irregular menstrual cycles and ovulatory dysfunction, and (3) polycystic ovaries observed on ultrasound. Notably, the 2023 guideline allows ultrasound to be replaced by anti-Müllerian hormone (AMH) levels in adults. Additionally, if both hyperandrogenism and ovulatory dysfunction are present, neither ultrasound nor AMH testing is necessary. For adolescents, PCOS can be diagnosed based on the presence of both hyperandrogenism and ovulatory dysfunction, without the need for ultrasound or AMH testing. Furthermore, regarding clinical hyperandrogenism, the 2023 guideline suggests that the presence of hirsutism alone should be considered predictive of biochemical hyperandrogenism and PCOS in adults. This streamlined approach is highly practical, offering greater convenience and lower costs for both clinicians and patients (Fig. 1).

The guideline emphasizes the need to acknowledge a wide range of comorbidities associated with PCOS, which include metabolic risk factors such as insulin resistance, diabetes mellitus, cardiovascular disease, dermatologic conditions, sleep disorders, and psychological issues. A significant number of individuals with PCOS exhibit metabolic abnormalities, notably insulin resistance and compensatory hyperinsulinemia [6]. How-

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Corresponding author: A Ram Hong

Division of Endocrinology and Metabolism, Department of Internal Medicine, Chonnam National University Medical School, 160 Baekseo-ro, Dong-gu, Gwangju 61469, Korea

Tel: +82-61-379-7822, Fax: +82-61-379-7628, E-mail: wanilove23@nate.com

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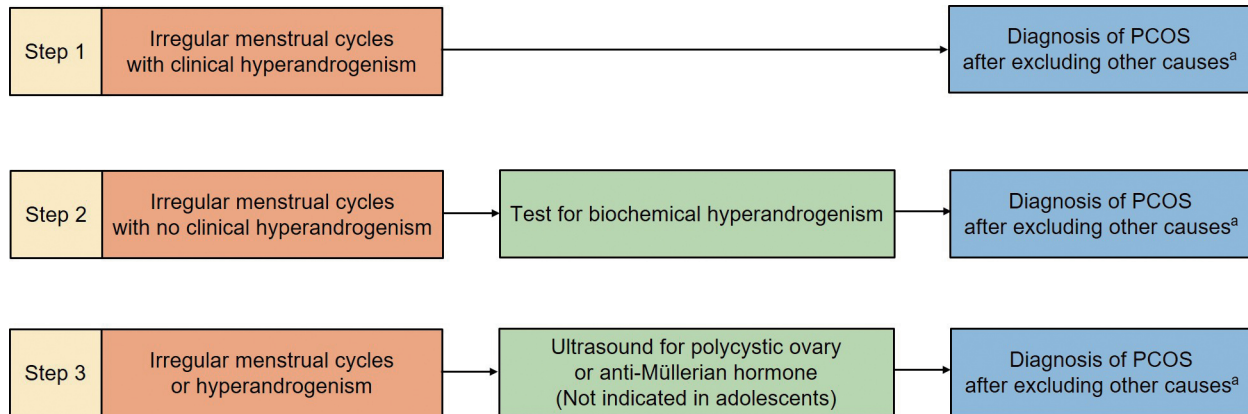


Fig. 1. Updated diagnostic algorithm based on the 2023 guideline for polycystic ovary syndrome (PCOS). ^aAt all stages of diagnosing PCOS, other potential causes (e.g., Cushing's syndrome, adrenal tumors) should be excluded using biochemical parameters such as thyroid-stimulating hormone, prolactin, 17-hydroxyprogesterone, follicle-stimulating hormone, or luteinizing hormone. For adolescents, PCOS can be diagnosed based on the presence of both hyperandrogenism and ovulatory dysfunction, without requiring ultrasound or anti-Müllerian hormone testing.

ever, it is not recommended to clinically measure insulin resistance due to the limited clinical relevance of current insulin assays. Psychological issues, including depression, anxiety, body image concerns, and eating disorders, are frequently overlooked but have a substantial impact on women of reproductive age and require careful attention. Additionally, women with PCOS face a higher risk of adverse pregnancy outcomes and endometrial cancer. Given the lifelong impact of PCOS from adolescence to the postmenopausal years, a comprehensive approach to long-term management and follow-up is strongly advised.

Alongside lifestyle interventions such as dietary changes and increased physical activity, weight management remains the most effective treatment for PCOS. Nonetheless, weight bias and stigma should be minimized. Prioritizing emotional well-being and quality of life is equally important. For pharmacological treatment, combined oral contraceptive pills are the first-line therapy for managing menstrual irregularities and hyperandrogenism. Metformin is the agent of choice for addressing metabolic derangements. Anti-obesity medications and bariatric surgery may also be considered, weighing their risks and benefits. For fertility treatment, letrozole is the first-line medication, followed by gonadotropins or ovarian surgery as second-line options. *In vitro* fertilization can be considered as a third-line therapy.

To enhance health outcomes and improve the disease burden of PCOS, it is crucial to prioritize education and awareness among women and healthcare professionals. This effort should be supported by advanced research on PCOS and the develop-

ment of evidence-based resources. A comprehensive understanding of the condition, combined with a high level of clinical vigilance, is essential for facilitating an accurate diagnosis and initiating effective management. According to the 2023 guideline, after diagnosis, it is important to adopt an integrated care model and engage in shared decision-making to improve the management and support provided to women with PCOS.

CONFLICTS OF INTEREST

No potential conflict of interest relevant to this article was reported.

ORCID

A Ram Hong <https://orcid.org/0000-0002-2494-9902>

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